

Welcome to STA 214

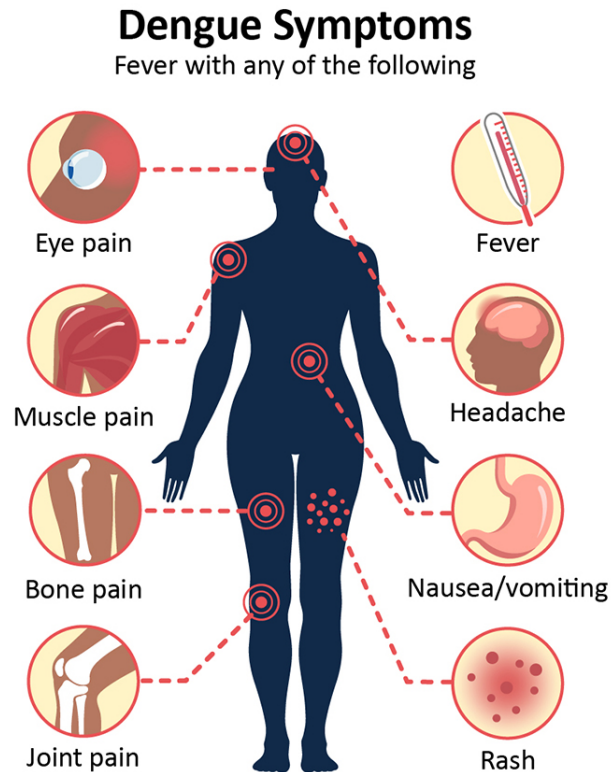
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Agenda

- + Introductions
- + Beginning logistic regression
- + Plan for week 1 and the semester
- + Syllabus highlights

Motivating example: Dengue fever

Dengue fever: a mosquito-borne viral disease affecting 400 million people a year



Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- + *Sex*: patient's sex (female or male)
- + *Age*: patient's age (in years)
- + *WBC*: white blood cell count
- + *PLT*: platelet count
- + other diagnostic variables...
- + *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

Warmup activity

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Research questions

- + How well can we predict whether a patient has dengue?
- + Which diagnostic measurements are most useful?
- + Is there a significant relationship between age and dengue?

How might you try to address each of these questions?

- regression model! (logistic regression)
↳ binary response
- hypothesis testing for regression coefficients
- make predictions with our model,
compare predictions to observed outcome

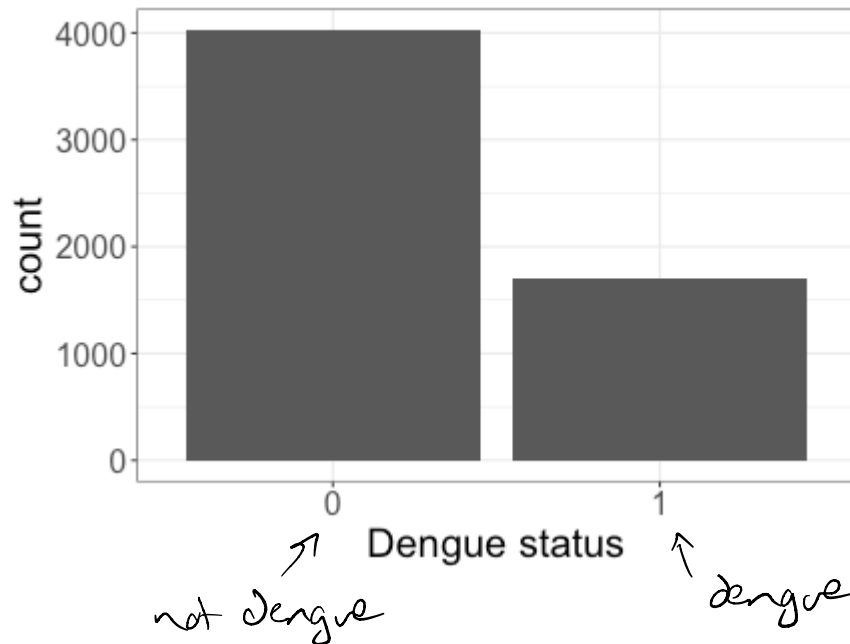
Exploratory data analysis (EDA)

What plot can we use to visualize the response (dengue status)?

Exploratory data analysis (EDA)

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Answer: Bar chart



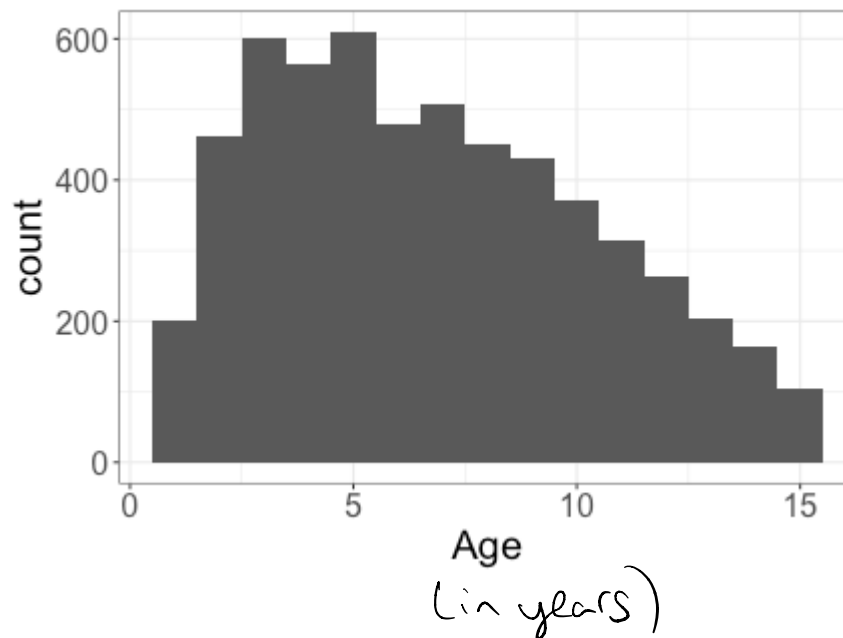
Exploratory data analysis (EDA)

What plot can we use to visualize an explanatory variable like Age?

Exploratory data analysis (EDA)

What plot can we use to visualize an explanatory variable like Age?

Answer: Histogram (or boxplot)



skewed right
peak around
4-5 years

Class activity

In the class activity (handout), you'll start to explore the relationship between age and dengue.

Spend a few minutes to do the following:

- + Say hi to the people around you, introduce yourself
- + Work in groups on the class activity
- + I will collect the handout at the end of class

Class activity

What do you think are some of the potential risks and benefits of the dengue study?

- Consent: parents have to give informed consent for children to participate
- Potential risks:
 - additional stress (may need to undergo additional tests)
- Benefits: better diagnosis of dengue fever (beneficial to other patients, not necessarily this sample)

Class activity

What is the (empirical) probability that a patient in the study has dengue?

$$P(\text{Dengue} = 1) = \frac{1697}{5720} \approx 0.3$$

Class activity

What is the (empirical) probability that a 5 year old patient has dengue? What about a 10 year old patient?

$$P(\text{Dengue} = 1 | \text{Age} = 5) = \frac{122}{122 + 487} \approx 0.2$$

$$P(\text{Dengue} = 1 | \text{Age} = 10) = \frac{185}{185 + 185} = 0.5$$

Odds

What are the (empirical) odds that a 5 year old patient has dengue?

$$\text{odds} = \frac{\pi}{1 - \pi} \quad \pi = \text{probability}$$

$$\text{5 year old patient:} \quad \text{odds} = \frac{0.2}{1 - 0.2} = \frac{0.2}{0.8}$$

$$= \frac{P(\text{Dengue} \mid \text{Age}=5)}{P(\text{no dengue} \mid \text{Age}=5)} = 0.25$$

$$P(\text{no dengue} \mid \text{Age}=5) = 4 \cdot P(\text{Dengue} \mid \text{Age}=5)$$

iid ; "independent and identically distributed"

Fitting a model: linear regression?

Response variable: Y_i = dengue status of i th patient

Explanatory variable: Age_i = dengue status of i th patient

What would a *linear* regression model look like for these variables?

$$Y_i = \underbrace{\beta_0 + \beta_1 Age_i}_{\text{line}} + \epsilon_i$$

intercept slope

Typically assume : $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

↑ ↑ ↑
"distributed" normal mean 0

error aka noise constant variance

Fitting a model: initial attempt

What if we try a linear regression model?

Y_i = dengue status of i th patient

$$Y_i = \beta_0 + \beta_1 \text{Age}_i + \varepsilon_i \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

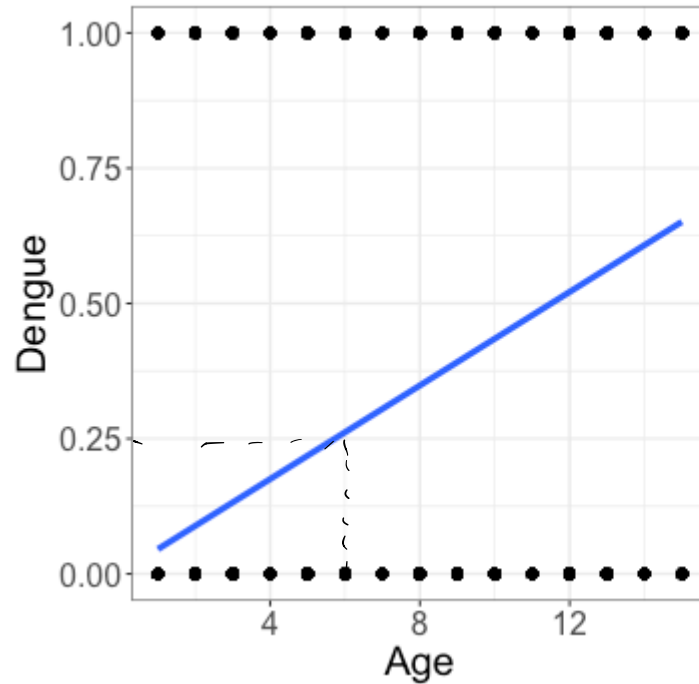
What are some potential issues with this linear regression model? Discuss with your neighbors for 1--2 minutes, then we will discuss as a class.

Y_i is binary (0 or 1)

$\beta_0 + \beta_1 \text{Age}_i + \varepsilon_i$ is continuous

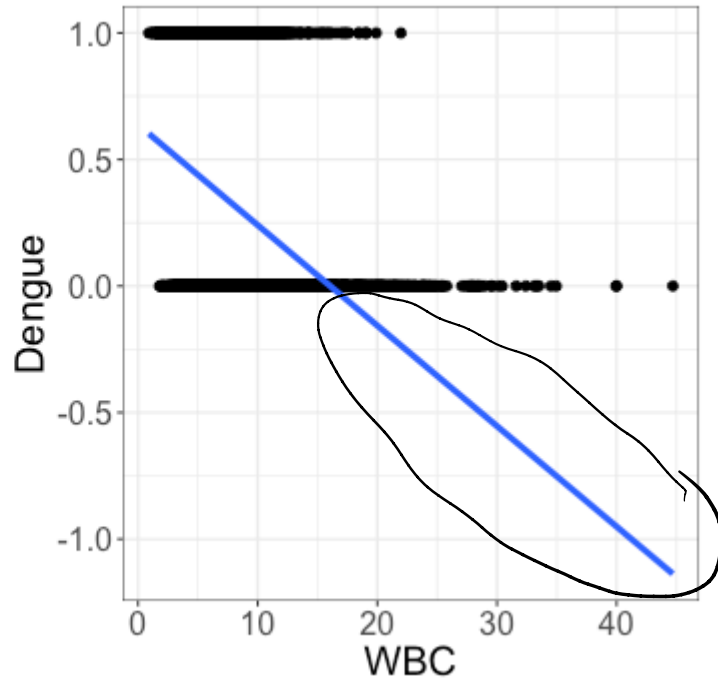
and doesn't have to be bounded

Don't fit linear regression with a binary response



$$\text{Age} = 6 \Rightarrow \hat{\text{Dengue}} = 0.25$$

?



$$\text{WBC} > 15 \\ \Rightarrow \hat{\text{Dengue}} < 0$$

??

Next steps

- + We will spend the next few weeks talking in depth about logistic regression
- + Download R and RStudio today or tomorrow
 - + Instructions on course website
 - + Please contact me if you have problems!
- + Bring laptop to class
- + HW 1 released on the course website
- + Course codebook also on the course website

Semester plan (tentative)

- + Logistic regression
- + Poisson regression + modifications
- + Mixed effects
- + Parametric bootstrapping (time permitting)

Course goals:

- + Be able to model data with different types of response variables
- + Extend tools from 112 to address violations of the usual linear regression assumptions
- + Emphasize sound statistical and data analysis practices

Course prerequisites

Prerequisites:

+ STA 112 and MTH 111 (Calculus I)

I expect you to be familiar with:

- + EDA (Exploratory data analysis)
- + Linear regression with multiple predictors
- + Comparing and interpreting models
- + Confidence intervals and hypothesis tests
- + Basic R computing

Expectations

- + Complete any assigned reading ahead of class
- + Bring laptop each day
- + Submit class activities (graded for effort, not completeness or correctness)
- + Attend department seminars (more info to follow)

Course components

- + Class participation (graded for effort)
- + HW assignments (roughly one per week)
- + Exams (2 midterms, 1 final)
- + Project

AI policy

- + I will *never* use AI to grade your work; all feedback you receive will be directly from me and the TA
- + Collaboration with other students, and AI assistance, is permitted on homework
 - + Assistance does *not* mean uploading the assignment to ChatGPT and copying the answers
 - + You must cite collaborators and external resources
- + See syllabus for further details

For next time

- + Make sure R and RStudio are installed
- + Instructions are provided on the course website
- + **Reading:** sections 3.3.1 and 6.2 in the textbook